



CRYSTAL CALIBRATION SALES AND SERVICE CO., LTD.

45/48 Soi Salathammasop31, Salathammasop Rd.,

Salathammasop, Thawewatthana, Bangkok 10170 Thailand

Tel : 0-2408-8474-5 Fax : 0-2408-8477 Email : info@crystalcal.com www.crystalcal.com



CERTIFICATE OF CALIBRATION

Certificate No. : 25-1390-025

Issue Date : 24 October 2025

Work Order No. : 25/1390

Customer Name : กลุ่มงานเทคนิคการแพทย์ โรงพยาบาลพระจอมเกล้า จังหวัดเพชรบุรี

53 ถ.รถไฟ ต.คลองกระแซง

อ.เมือง จ.เพชรบุรี 76000

Date of Received : 20 October 2025

Date of Calibration : 24 October 2025

Instrument Details : Description : Piston pipette (Auto pipette)
Manufacturer : DLAB
Model : N/A
Serial No. : YM196AH0057239
ID No. : ค18-ป001-01-2/63
Volume range : 2 μ l to 20 μ l
Type : Variable volume single channel (air displacement)
Location : Mass and Volumetric Calibration Laboratory

Calibration Method : This instrument was calibrated by Gravimetric methods for the determination of measurement error according to calibration procedure no. CWI-V-01 based on ISO 8655 : 2002 Part 2, 6

Environmental Condition

Ambient Temperature : Control at 20 °C $\pm$ 2.5 °C
Relative humidity : Control at 50 %RH $\pm$ 10 %RH
Atmospheric pressure : Monitoring at 1010 hPa $\pm$ 10 hPa

Traceability of Measurement

: This certificate of calibration documents the traceability to national standard, which realize the unit of measurement according to the International system of Units (SI)

Calibrated by : Miss Nattharika Peedee
Calibration Engineer

Approved by :

(Mr. Anuwat Yaklermjit)
Laboratory Manager

This certificate may not be reproduced other than in full except with the prior written approval of Crystal Calibration Sales and Service co., Ltd.

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Details of Calibration

1. Reference Standards Instrument

Instrument	Serial No. / ID No.	Certification	Due Date
1.1 Electronic Balance	BAL-06	25-0164-004	14 March 2026
1.2 Digital Thermometer with sensor	RTD-09	25-0164-009	3 March 2026
1.3 Thermo hygrometer	TH-06	25-0164-012	3 March 2026
1.4 Barometer	BM-06	L202509118-0001	9 September 2026

2. Certificate traceable : This certificate traceable to The International System of Unit refer to

1.1 Crystal Calibration Sales and Service Co., Ltd., NSC-ONSC Calibration No. 0260

1.2 to 1.3 Crystal Calibration Sales and Service Co., Ltd., NSC-ONSC Calibration No. 0260

1.4 Miracle international Technology Co., Ltd., NSC-ONSC Calibration No. 0052

3. Condition of item : Used

4. Calibration site : Permanent

Result of Calibration

This result of calibration : Before adjustment

Resolution : 0.5 μ l

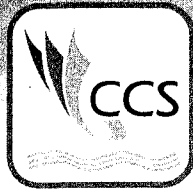
Test volume	(μ l)	2	10	20
Setting volume	(μ l)	2	10	20
Mean volume	(μ l)	2.1093	10.1361	20.2693
Uncertainty of measurement	($\pm \mu$ l)	0.030	0.038	0.064
Coverage factor (k)		2.23	2.28	2.28
Systematic : Error (e_s)	(μ l)	0.1093	0.1361	0.2693
: Error in percent (% e_s)		0.546	0.680	1.346
Random : Error (S_r)	(μ l)	0.039	0.050	0.054
: Error in percent (%CV)		1.860	0.495	0.266

Note :

- Nominal volume piston pipette : 20 μ l
- The result of repeat the test 10 measurements
- The Calibration process calcurate temperature reference is 20°C
- The values obtained are the mass values of the volume delivered (Ex.)
- Pipette tip : Manufacturer: BIOHIT ; Color code: Colorless ; Capacity: 0.5 ul to 300 ul ; Length : 50 mm ; None filter
- The standard uncertainty of measurement has been determined in accordance with ISO/TR 20461:2000
- Calibrate items it good condition this report customer request and accepted in certificate

This result of calibration was found accurate as shown on date and place of calibration only.

The reported uncertainty of measurement was based on a standard uncertainty multiplied by a coverage factor $k=2$ providing a level of confidence of approximately 95%.

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1.4 Miracle international Technology Co., Ltd., NSC-ONSC Calibration No. 0052

3. Condition of item : Used**4. Calibration site** : Permanent**Result of Calibration**

This result of calibration : After adjustment

Resolution : 0.5 μ l

Test volume	(μ l)	2	10	20
Setting volume	(μ l)	2	10	20
Mean volume	(μ l)	1.8947	9.9397	19.9981
Uncertainty of measurement	($\pm$ μ l)	0.027	0.036	0.064
Coverage factor (k)		2.21	2.25	2.28
Systematic : Error (e_s)	(μ l)	-0.1053	-0.0603	-0.0019
: Error in percent (% e_s)		-0.527	-0.301	-0.009
Random : Error (S_r)	(μ l)	0.035	0.047	0.052
: Error in percent (%CV)		1.850	0.473	0.262

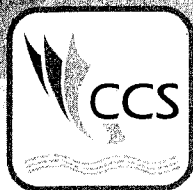
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- Calibrate items it good condition this report customer request and accepted in certificate

This result of calibration was found accurate as shown on date and place of calibration only.

The reported uncertainty of measurement was based on a standard uncertainty multiplied by a coverage factor $k=2$ providing a level of confidence of approximately 95%.

-END-



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Statements of conformity report

Refer to Certificate No. : 25-1390-025

Issue Date : 24 October 2025

Work Order No. : 25/1390

Detail of Equipment

Description : Piston pipette (Auto pipette)

Manufacturer : DLAB

ID No. : ๙18-๗001-01-2/63

Model : N/A

Volume range : 2 μ l to 20 μ l

Serial No. : YM196AH0057239

Result of Calibration :

This result of calibration : After adjustment

Resolution : 0.5 μ l

Test volume	(μ l)	2	10	20	
Setting volume	(μ l)	2	10	20	
Mean volume	(μ l)	1.8947	9.9397	19.9981	
Uncertainty of measurement	($\pm \mu$ l)	0.027	0.036	0.064	
Coverage factor (k)		2.21	2.25	2.28	
Systematic : Error (e_s)	(μ l)	-0.1053	-0.0603	-0.0019	
: Error in percent (% e_s)		-5.265	-0.603	-0.009	
Random : Error (S_r)	(μ l)	0.035	0.047	0.052	
: Error in percent (%CV)		1.752	0.470	0.262	
Evaluate report (Systematic)	: Abs. Error+Uncertainty	(μ l)	0.1323	0.0963	0.0659
	: MPE. Error (e_s)	(μ l)	0.2	0.2	0.2
	: Result (PASS / FAIL)		<u>PASS</u>	<u>PASS</u>	<u>PASS</u>
	: Abs. Error+Uncertainty in percent		6.6903	0.9652	0.3293
	: MPE. Error in percent (% e_s)		10	2	1
	: Result (PASS / FAIL)		<u>PASS</u>	<u>PASS</u>	<u>PASS</u>
Evaluate report (Random)	: MPE Error (S_r)	(μ l)	0.1	0.1	0.1
	: Result (PASS / FAIL)		<u>PASS</u>	<u>PASS</u>	<u>PASS</u>
	: MPE Error in percent (CV)		5	1	0.5
	: Result (PASS / FAIL)		<u>PASS</u>	<u>PASS</u>	<u>PASS</u>

This statement judgment and decision by Laboratory guideline only

The conformity certificate documents validity following ISO/IEC Guide 98-4 : Role of measurement uncertainty in conformity assessment based on statement with guard band refer to ISO 8655 : 2022 consider expanded measurement uncertainty (k=95%)

The tolerance and decision rules ;

MPE of Systematic error = Measurement uncertainty + Absolute error ; $\leq$ MPE is pass , $>$ MPE is Fail

MPE of Random error = Random error comparative with ISO 8655 : 2022 in table 1 ; $\leq$ MPE is pass , $>$ MPE is Fail

Statements of conformity decision by :

(Mr. Anuwat Yaklermjit)

Laboratory Manager